📘 What is a Percentile?

A **percentile** is a measure that indicates the value below which a given percentage of observations in a dataset fall.

✅ In Simple Words:

- The **nth percentile** means **n percent of the data** is **less than or equal to** that value.
- Example: If your score is in the **90th percentile**, then **90%** of people scored **less than or equal** to you.

📌 Percentiles vs. Quartiles

Term	Meaning
25th percentile	Q1 (First Quartile)
50th percentile	Q2 (Median)
75th percentile	Q3 (Third Quartile)

📊 Percentile Example with Step-by-Step Calculation

Let's say we have the following **ordered** data:

Data: 5, 7, 10, 12, 14, 15, 18, 21, 25, 30

(10 values)

🎯 Goal: Find the 70th percentile.

Step 1: Use the Percentile Formula

P_k represents the kth percentile and its position/rank in the dataset is calculated as:

$$r = \lfloor \frac{k}{100} \cdot (n+1) \rfloor$$

Where:

- n = total number of values = 10
- k = desired percentile = 70

$$r = \frac{10+1}{100} \times 70$$

$$r = 7.7$$

Since rank/position r is not integer number, so, the 70th percentile lies **between the 7th and 8th values**.

Step 2: Interpolate Between Values

7th value = 18
8th value = 21

$$P_{70} = 18 + 0.7 \times (21 - 18) = 18 + 0.7 \times 3 = 18 + 2.1 = 20.1$$

✅ **The 70th percentile is 20.1**

📈 **Interpreting Percentiles**

Suppose a student scored **85** in a test and it corresponds to the **90th percentile**:

- This means **90%** of students scored **85 or less**.
- Only **10%** of students scored **higher** than 85.

🔁 **Other Examples:**

Percentile	Meaning
10th	10% of values are below this point
25th (Q1)	First quartile
50th (Q2)	Median (middle of the data)
75th (Q3)	Third quartile
90th	90% of data falls below this value

Rea-Life Examples:

Scenario	Interpretation

Your SAT score is in the **90th percentile**	You scored better than 90% of test takers
25th percentile of income = $25,000	25% of people earn less than $25,000
50th percentile of height = 5'9"	50% of people are shorter than 5'9" (i.e., median)

To determine whether a value like **85 is at the 90th percentile**, you need a **dataset** and a method to compute **percentile rank**.

✅ 1. Percentile Rank Formula

Given a value x, the **percentile rank** tells what percentage of data is **less than or equal to x**.

$$\text{Percentile Rank of } x = \frac{\text{Number of values } \leq x}{\text{Total number of values}} \times 100$$

□ Example

Suppose you have these exam scores:

Scores: 45, 50, 60, 65, 70, 75, 80, 82, 85, 90, 95

There are 11 values.

What percentile is the score 85?

☞ Count how many scores are **less than or equal to 85**:

45, 50, 60, 65, 70, 75, 80, 82, 85 → 9 values

Apply the formula:

$$\text{Percentile Rank of } 85 = \frac{9}{11} \times 100 = 81.8$$

✅ So, **85 is at the ~82nd percentile**, **not 90th**.

📌 So How Would You Know If 85 is in the 90th Percentile?

You can **only know** that **after** comparing it to the full dataset. You:

- **Sort the data**
- **Count how many values are below or equal to 85**
- **Compute the percentile rank**

 Important Notes:0

- A **percentile is always relative** to a dataset.
- Without the dataset, you **cannot** say whether 85 is in the 90th percentile.
- You might see statements like "85 is in the 90th percentile" from **standardized test reports** — in those cases, the percentile is **pre-computed** based on a **national or known distribution**.

Percentile calculation can be done in **different ways** depending on the method or tool used (e.g., Python, Excel, statistics books, or software like R). These methods vary in **how they interpolate** between data points.

🎯 Why So Many Methods?

When the desired percentile doesn't fall exactly on a data point (e.g., the 70th percentile in a 10-value dataset), different methods use different rules to estimate the value.

✅ Common Percentile Calculation Methods

1. Rank-Based Formula (Linear Interpolation)

This is a widely used method.

$$P_k = x_{(r)} + d \cdot (x_{(r+1)} - x_{(r)})$$

Where:

- P_k = k-th percentile
- $r = \lfloor \frac{k}{100} \cdot (n+1) \rfloor$
- d = fractional part (e.g., if rank = 4.6, then d = 0.6)
- $x_{(r)}$ and $x_{(r+1)}$ are values at ranks r and $r+1$

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2. Nearest Rank Method (Used by Excel & simple textbooks)

$$P_k = x_{(\lceil \frac{k}{100} \cdot n \rceil)}$$

- Round **up** to the nearest rank
- No interpolation — just pick the value at that position

✅ **Simple but can be imprecise**, especially for small datasets

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3. Weighted Average (Used in Python - NumPy)

NumPy offers multiple methods via np.percentile() with method= or interpolation= options (formerly):

Method	Description
linear	Linear interpolation between closest ranks
lower	Always pick the lower rank
higher	Always pick the higher rank
nearest	Pick the nearest data point
midpoint	Average of lower and higher

Example in Python:

```
import numpy as np
data = [10, 20, 30, 40, 50]
np.percentile(data, 40, method='linear')    # 40th percentile
np.percentile(data, 40, method='nearest')   # different result
```

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4. R's Type 1–9 Percentiles (Used in quantile() function)

R defines **9 different types** of percentile calculations:

- Type 1: Nearest rank
- Type 2–9: Different interpolation schemes

Python's pandas and scipy.stats libraries also support some of these via parameters.

✅ Summary Table

Method	Interpolation	Used In	Precision
Nearest Rank	❌ No	Excel (PERCENTILE.INC)	Simple
Linear Interp.	✅ Yes	NumPy, R, textbooks	Accurate
Midpoint	✅ Yes	NumPy (method="midpoint")	Medium
R Type 7	✅ Yes	Default in R and NumPy	Common

📌 Real-Life Example

Let's say we have this dataset:

Data = [15, 20, 35, 40, 50]

Find **40th percentile**:

Method	Result
Nearest	20
Linear	29.0
Midpoint	27.5

💡 Which Method Should You Use?

- **Large datasets**: Any method gives similar results
- **Small datasets**: Be aware of the method (linear is better)
- **In Python**: Use np.percentile(data, k, method='linear') or similar

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A **structured interpretation** of percentiles using a **student score dataset**, without showing computations:

🎓 Scenario: Student Exam Scores

We have the test scores of **10 students**:

[55, 60, 65, 68, 70, 72, 75, 78, 85, 95]

This is a **realistic, sorted dataset** representing student performance in a standardized test.

📊 Percentile-Based Interpretation

◆ 25th Percentile (Q1) – Lower Performers

- Represents the **score below which 25% of students** scored.
- Interpretation:

  One-fourth of the class scored **below approximately 64**.
  These students may need **extra support or remediation**.

◆ 50th Percentile (Median) – Middle Performers

- The **middle score** in the distribution.
- Interpretation:

  Half the students scored **below approximately 71** and half scored above.
  This gives a **central benchmark** for average performance.

◆ 75th Percentile (Q3) – Higher Performers

- Indicates the score **below which 75% of students** fall.
- Interpretation:

Three-fourths of the students scored **below approximately 78**.
The top 25% of students scored **above 78**, suggesting **high achievement**.

◆ **90th Percentile – Top Performers**

- Represents the **cutoff for the top 10%** of the class.
- Interpretation:

  A student scoring above approximately **94** is in the **top 10%**.
  These students may benefit from **advanced learning opportunities**.

📌 Educational Implications

Percentile	Insight	Action
25th	Identifies students struggling the most	Provide support, tutoring, or intervention
50th	Indicates average performance	Monitor for general trends
75th	Highlights high achievers	Encourage enrichment activities
90th	Flags exceptional students	Recommend honors or advanced programs

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✅ Summary

Percentiles provide a **relative understanding of student performance**:

- They help **categorize students** into performance bands.
- Enable **targeted educational decisions** rather than treating the class as a uniform group.
- Are useful for **benchmarking** and **tracking improvement** over time.
